



NORTH SYDNEY BOYS' HIGH SCHOOL
2009 Year 11 Assessment Task 2

MATHEMATICS

(Extension 1)

General instructions

- Working time – 50 minutes.
- Write on your own A4 paper.
Each question is to commence on a new page.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question (Insufficient/illegible working may cause a deduction of marks).
- Attempt **all** questions.

Class teacher (please ✓)

- ☐ Mr Barrett
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Ireland
- ☐ Mr Weiss
- ☐ Mr Lam

NAME: PAGES USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{9}$	$\overline{13}$	$\overline{18}$	$\overline{11}$	$\overline{17}$	$\overline{68}$	

Question 1 (9 Marks)

Commence a NEW page.

Marks

Solve, giving exact values:

- (a) $x + 6 = 3 - x$. **1**
- (b) $\frac{2}{x+5} - \frac{3}{x-5} = \frac{1}{x^2-25}$. **2**
- (c) $(x+1)^2 = 8$. **2**
- (d) $x(x+1) = 1$. **2**
- (e) $|3x+2| = 1-x$. **2**

Question 2 (13 Marks)

Commence a NEW page.

Marks

- (a) Write the exact values of
- $\cos 180^\circ$. **1**
 - $\cot 240^\circ$. **1**
 - $\tan 315^\circ$. **1**
- (b)
- Write 75° in radians as an exact value. **1**
 - Evaluate $\cos^2 2.3$ correct to 3 decimal places. **2**
- (c) Find the value of the pronumeral in each of the following diagrams.



- (d) A ship leaves port A on a bearing of 120°T travelling for 2 hours at 30 knots (nautical miles/hr). It changes to a new bearing of 220°T for 1 hour at 20 knots. **3**

How far is it from port A ?

Question 3 (18 Marks)

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Marks

(a) Solve the following inequalities:

i. $\frac{3x}{4} \leq \frac{7x}{3} - 2.$

2

ii. $x^2 + x > 2.$

2

iii. $|2x + 1| < 3.$

3

iv. $\frac{x+2}{2x-3} \leq 1.$

3

(b) Form an equation and solve:

i. There are 450 students at a particular school. If there are 40 more boys than girls, how many boys and girls are there?

2

ii. A box contains 50 coins made up of 10 cent coins and 20 cent coins. How many of each are in the box if the value of the coins is \$7?

3

(c) Find the coordinates of the points of intersection of the graphs

3

$$y = x - 3 \quad \& \quad x^2 + y^2 - 4y - 13 = 0$$

Question 4 (11 Marks)

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Marks(a) If $\tan \theta = -\frac{4}{3}$ and $90^\circ < \theta < 180^\circ$, find the exact value of

i. $\sin \theta.$

2

ii. $\cot \theta.$

1

(b) Simplify fully

i. $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}.$

2

ii. $1 - \sin^2(180^\circ - \theta).$

2

(c) Prove the following identities:

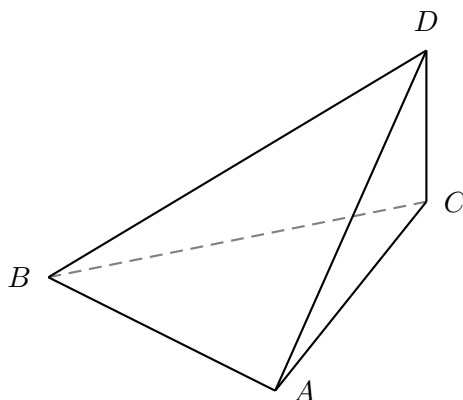
i. $\tan \theta (1 - \cot^2 \theta) + \cot \theta (1 - \tan^2 \theta) = 0.$

2

ii. $\sec \theta + \tan \theta = \frac{1 + \sin \theta}{\cos \theta}.$

2

- | Question 5 (17 Marks) | Commence a NEW page. | Marks |
|--|----------------------|-------|
| (a) Consider the function $y = 4 \sin 3x$, where x is in degrees. | | |
| i. Find the period. | | 1 |
| ii. Find the amplitude. | | 1 |
| iii. Sketch the function for $0^\circ \leq x \leq 360^\circ$. | | 2 |
| (b) Solve for $0^\circ \leq x \leq 360^\circ$ | | |
| i. $\cos x = -\frac{1}{2}$. | | 2 |
| ii. $\tan 2x = \frac{1}{\sqrt{3}}$ | | 3 |
| iii. $3 \cos^2 x + 5 \sin x = 1$. | | 4 |
| (c) The points A and B are 500 m apart on the ground and D is the top of a tower. | | |
| $\angle BAD$ and $\angle DBA$ are 59° and 54° respectively. The elevation of D from A is 5° . | | |



Copy the diagram into your writing booklet and mark on the figure all the angles stated above.

- | | |
|---|---|
| i. Show that the height h metres of the tower is given by | 3 |
|---|---|

$$h = \frac{500 \sin 5^\circ \sin 54^\circ}{\sin 67^\circ}$$

- | | |
|------------------------------------|---|
| ii. Find h to the nearest metre. | 1 |
|------------------------------------|---|

End of paper.

Suggested Solutions

Question 1

(a) (1 mark)

$$x + 6 = 3 - x$$

$$2x = -3$$

$$x = -\frac{3}{2}$$

(b) (2 marks)

✓ [1] for removing denominators.

✓ [1] for $x = -26$.

$$\frac{2}{x+5} - \frac{3}{x-5} = \frac{1}{x^2-25}$$

$$\times(x-5)(x+5) \quad \times(x-5)(x+5) \quad \times(x-5)(x+5)$$

$$2(x-5) - 3(x+5) = 1$$

$$2x - 10 - 3x - 15 = 1$$

$$-x - 25 = 1$$

$$x = -26$$

(c) (2 marks)

$$(x+1)^2 = 8$$

$$x+1 = \pm\sqrt{8}$$

$$x = -1 \pm \sqrt{8}$$

$$= -1 \pm 2\sqrt{2}$$

(d) (2 marks)

$$x^2 + x - 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$= \frac{-1 \pm \sqrt{5}}{2}$$

(e) (2 marks)

$$|3x+2| = 1-x$$

$$|3x+2| = \begin{cases} 3x+2 & 3x+2 \geq 0 \\ -(3x+2) & 3x+2 < 0 \end{cases}$$

$$\left. \begin{array}{l} 3x+2 = 1-x \\ 4x = -1 \\ x = -\frac{1}{4} \end{array} \right| \begin{array}{l} -(3x+2) = 1-x \\ -3x-2 = 1-x \\ -2x = 3 \\ x = -\frac{3}{2} \end{array}$$

Test values:

$$\bullet x = -\frac{1}{4}.$$

$$\left| 3\left(-\frac{1}{4}\right) + 2 \right| = \left| -\frac{3}{4} + 2 \right|$$

$$= \left| \frac{5}{4} \right| \checkmark$$

$$1 - \left(-\frac{1}{4}\right) = \frac{5}{4} \checkmark$$

$$\bullet x = -\frac{3}{2}.$$

$$\left| 3\left(-\frac{3}{2}\right) + 2 \right| = \left| -\frac{9}{2} + 2 \right|$$

$$= \left| -\frac{5}{2} \right| = \frac{5}{2} \checkmark$$

$$1 - \left(-\frac{3}{2}\right) = \frac{5}{2} \checkmark$$

Question 2

(a) i. (1 mark)

$$\cos 180^\circ = -1$$

ii. (1 mark)

$$\cot 240^\circ = \frac{1}{\tan(180^\circ + 60^\circ)} = \frac{1}{\sqrt{3}}$$

iii. (1 mark)

$$\tan 315^\circ = \tan(360^\circ - 45^\circ) = -1$$

(b) i. (1 mark)

$$75^\circ \Rightarrow \frac{75^\circ}{180^\circ} \times \pi = \frac{5\pi}{12}$$

ii. (2 marks)

$$(\cos 2.3)^2 = 0.444$$

(c) i. (2 marks)

$$\frac{x}{4} = \sin 59^\circ$$

$$x = 4 \sin 59^\circ = 3.43 \text{ (2 d.p.)}$$

ii. (2 marks)

$$\cos \theta = \frac{b^2 + c^2 - a^2}{2bc}$$

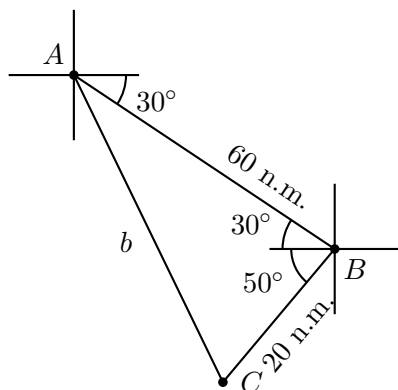
$$= \frac{4^2 + 9^2 - 6^2}{2 \times 4 \times 9}$$

$$= \frac{61}{72}$$

$$\theta = 32^\circ 5'$$

(d) (3 marks)

- ✓ [1] for correct diagram.
- ✓ [1] for correct application of cosine rule.
- ✓ [1] for $b = 59.9$ to 1 d.p.



$$\begin{aligned}
 b^2 &= a^2 + c^2 - 2ac \cos B \\
 &= 3600 + 400 - 2400 \cos 80 \\
 &= 3583.24 \\
 \therefore b &= 59.9 \text{ n.m. (1 d.p.)}
 \end{aligned}$$

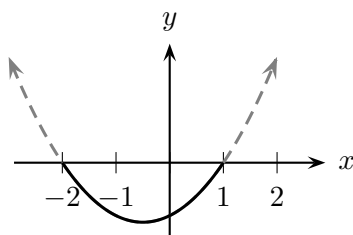
Question 3

(a) i. (2 marks)

$$\begin{aligned}
 \frac{3x}{4} &\leq \frac{7x}{3} - 2 \\
 9x &\leq 28x - 24 \\
 24 &\leq 19x \\
 x &\geq \frac{24}{19}
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 x^2 + x - 2 &> 0 \\
 (x + 2)(x - 1) &> 0
 \end{aligned}$$



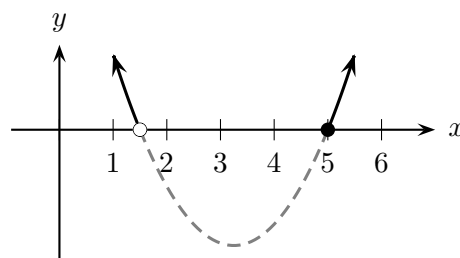
$$\therefore x < -2 \text{ or } x > 1.$$

iii. (3 marks)

$$\begin{aligned}
 |2x + 1| &< 3 \\
 |2x + 1| &= \begin{cases} 2x + 1 & 2x + 1 \geq 0 \\ -(2x + 1) & 2x + 1 < 0 \end{cases} \\
 2x + 1 < 3 & \quad | \quad -(2x + 1) < 3 \\
 2x < 2 & \quad | \quad -2x < 4 \\
 x < 1 & \quad | \quad x > -2 \\
 \therefore -2 < x < 1
 \end{aligned}$$

iv. (3 marks)

$$\begin{aligned}
 \frac{x+2}{2x-3} &\leq \frac{1}{x(2x-3)^2} \\
 (x+2)(2x-3) &\leq (2x-3)^2 \\
 (2x-3)^2 - (x+2)(2x-3) &\geq 0 \\
 (2x-3)((2x-3) - (x+2)) &\geq 0 \\
 (2x-3)(x-5) &\geq 0
 \end{aligned}$$



$$x < \frac{3}{2} \text{ or } x \geq 5$$

(b) i. (2 marks)

- ✓ [1] for solving equation, resulting in $x = 205$.
- ✓ [1] for conclusion.

Let the # of girls be x ,

$$\begin{aligned}
 (x + 40) + x &= 450 \\
 2x &= 410 \\
 x &= 205
 \end{aligned}$$

Hence there are 205 girls & 245 boys.

ii. (3 marks)

✓ [1] for setting up simultaneous equations.

✓ [1] for solving simultaneous equations.

✓ [1] for correct final conclusion.

Let the # of 10c coins be x , and # of 20c coins be y .

$$\begin{cases} x + y = 50 & (1) \\ 10x + 20y = 700 & (2) \end{cases}$$

$$(2) - [(1) \times 10]$$

$$10y = 200$$

$$y = 20$$

$$\therefore x = 30$$

Hence there are thirty 10c coins and twenty 20c coins.

(c) (3 marks)

✓ [1] for correct substitution

✓ [1] for solution of $x = 1$, $x = 4$.

✓ [1] for correct points of intersection.

$$\begin{cases} y = x - 3 & (1) \\ x^2 + y^2 - 4y - 13 = 0 & (2) \end{cases}$$

Substitute (1) to (2)

$$x^2 + (x - 3)^2 - 4(x - 3) - 13 = 0$$

$$x^2 + x^2 - 6x + 9 - 4x + 12 - 13 = 0$$

$$2x^2 - 10x + 8 = 0$$

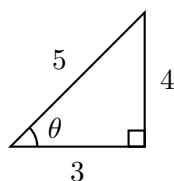
$$x^2 - 5x + 4 = 0$$

$$(x - 4)(x - 1) = 0$$

$$x = 1 \text{ or } 4$$

Points of intersection are $(1, -2)$ and $(4, 1)$.**Question 4**

(a) i. (2 marks)



$$\sin \theta = \frac{4}{5}$$

ii. (1 mark)

$$\cot \theta = -\frac{3}{4}$$

(b) i. (2 marks)

$$\begin{aligned} & \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \\ &= \frac{\sin^2 \theta + (1 + 2 \cos \theta + \cos^2 \theta)}{\sin \theta (1 + \cos \theta)} \\ &= \frac{2 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \\ &= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \\ &= \frac{2}{\sin \theta} = 2 \operatorname{cosec} \theta \end{aligned}$$

ii. (2 marks)

$$1 - \sin^2(180^\circ - \theta) = 1 - \sin^2 \theta = \cos^2 \theta$$

(c) i. (2 marks)

$$\begin{aligned} & \tan \theta (1 - \cot^2 \theta) + \cot \theta (1 - \tan^2 \theta) \\ &= \tan \theta - \frac{1}{\tan \theta} + \frac{1}{\tan \theta} - \tan \theta \\ &= 0 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} \sec \theta + \tan \theta &= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \\ &= \frac{1 + \sin \theta}{\cos \theta} \end{aligned}$$

Question 5

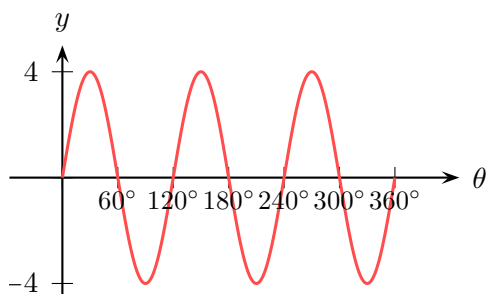
(a) i. (1 mark)

$$T = \frac{360^\circ}{3} = 120^\circ$$

ii. (1 mark)

$$a = 4$$

iii. (2 marks)



(b) i. (2 marks)

$$\cos x = -\frac{1}{2}$$

$$x = 120^\circ, 240^\circ$$

ii. (3 marks)

- ✓ [1] for $0 \leq 2x \leq 720^\circ$.
- ✓ [1] for finding the values of $2x$.
- ✓ [1] for finding the values of x .

$$0^\circ \leq 2x \leq 720^\circ$$

$$\tan 2x = \frac{1}{\sqrt{3}}$$

$$2x = 30^\circ, 210^\circ, 390^\circ, 570^\circ$$

$$\therefore x = 15^\circ, 105^\circ, 195^\circ, 285^\circ$$

iii. (4 marks)

- ✓ [1] for changing $\cos^2 x \rightarrow 1 - \sin^2 x$.
- ✓ [1] for rearranging to the quadratic $3 \sin^2 x - 5 \sin x - 2 = 0$.
- ✓ [1] for solving the quadratic.
- ✓ [1] for final answer.

$$3 \cos^2 x + 5 \sin x = 1$$

$$3(1 - \sin^2 x) + 5 \sin x = 1$$

$$3 - 3 \sin^2 x + 5 \sin x = 1$$

$$3 \sin^2 x - 5 \sin x - 2 = 0$$

Let $m = \sin x$,

$$3m^2 - 5m - 2 = 0$$

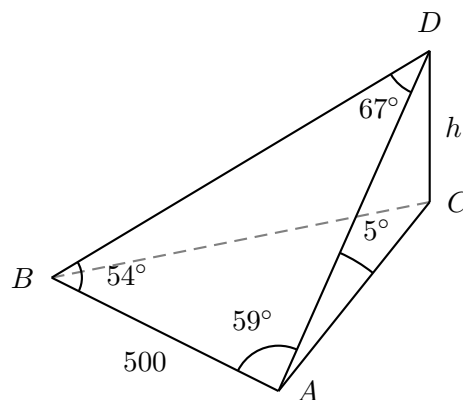
$$(3m + 1)(m - 2) = 0$$

$$\therefore m = -\frac{1}{3}, m = 2$$

$\therefore \sin x = -\frac{1}{3}$ as $\sin x = 2$ has no solution

$$\therefore x = 199^\circ 28' \text{ or } 340^\circ 32'$$

(c) i. (3 marks)



$$\frac{h}{AD} = \sin 5^\circ$$

$$\therefore AD = \frac{h}{\sin 5^\circ}$$

Applying the sine rule to $\triangle DBA$,

$$\frac{AD}{\sin 54^\circ} = \frac{500}{\sin 67^\circ}$$

$$\frac{h}{\sin 5^\circ \sin 54^\circ} = \frac{500}{\sin 67^\circ}$$

$$\therefore h = \frac{500 \sin 5^\circ \sin 54^\circ}{\sin 67^\circ}$$

ii. (1 mark)

$$h = 38.3 \text{ m} = 38 \text{ m (nearest m)}$$